

CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$. Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$ (columns: Accepted by $\mathcal{A}_1$? and Accepted by $\mathcal{A}_2$?).

Exercise 1.2

Describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba

3. All the words that contain an odd number of a 's and an odd number of b 's
4. All the words that contain an even number of a 's and an odd number of b 's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answers:

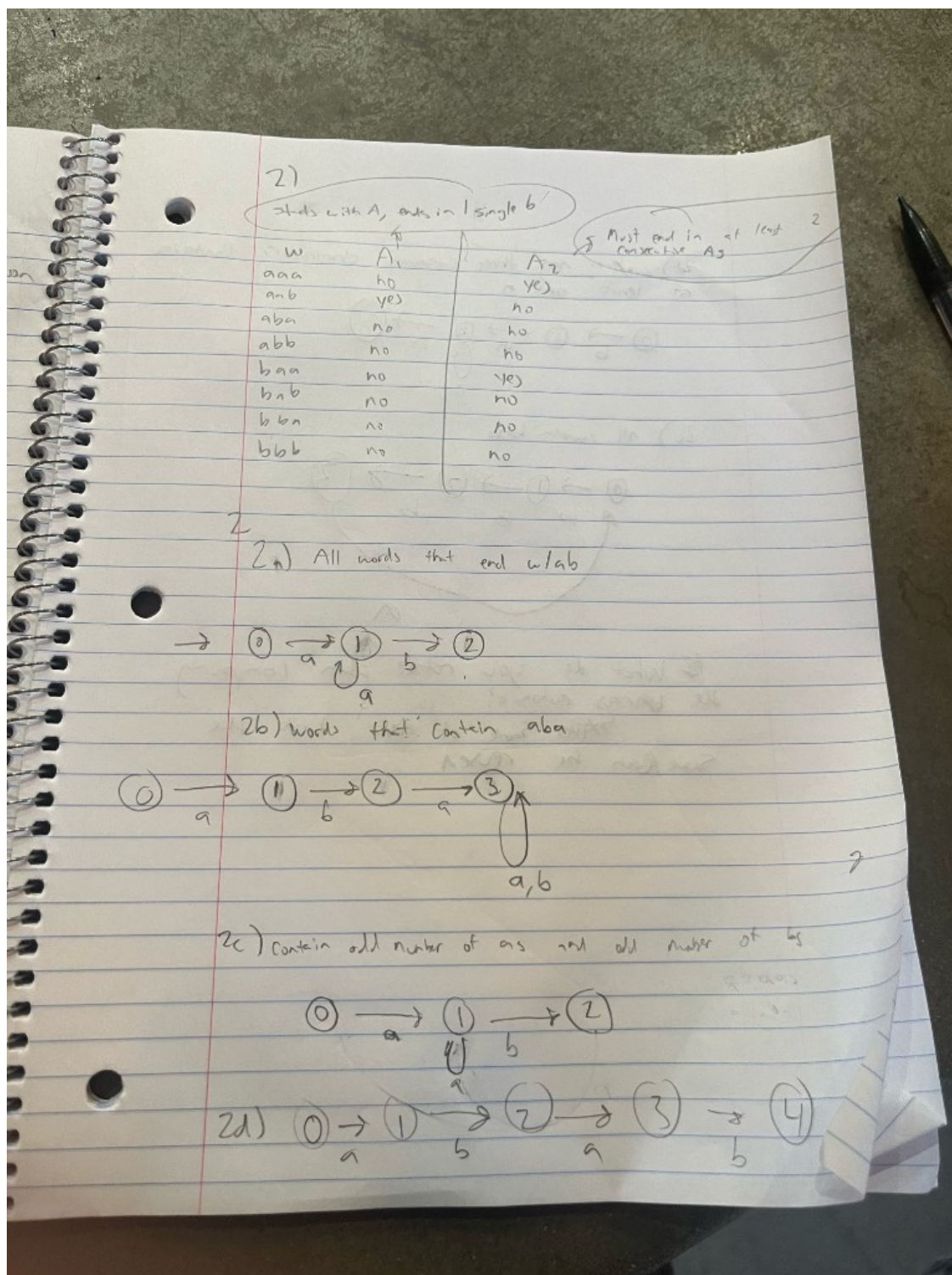
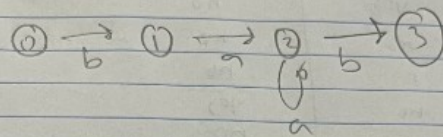
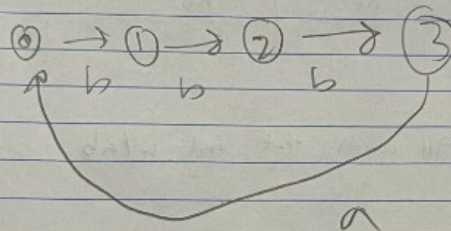


Figure 1: Week 1 answers (table, language conditions, DFA designs 2a-2d).

2d) any three consecutive characters contain at least one a



2c) all contain bbb



What do you notice when comparing the two automata?
 Some of them look the same/can be reused

Figure 2: Week 1 answers (DFA designs 2c, 2d; comparison of automata).

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

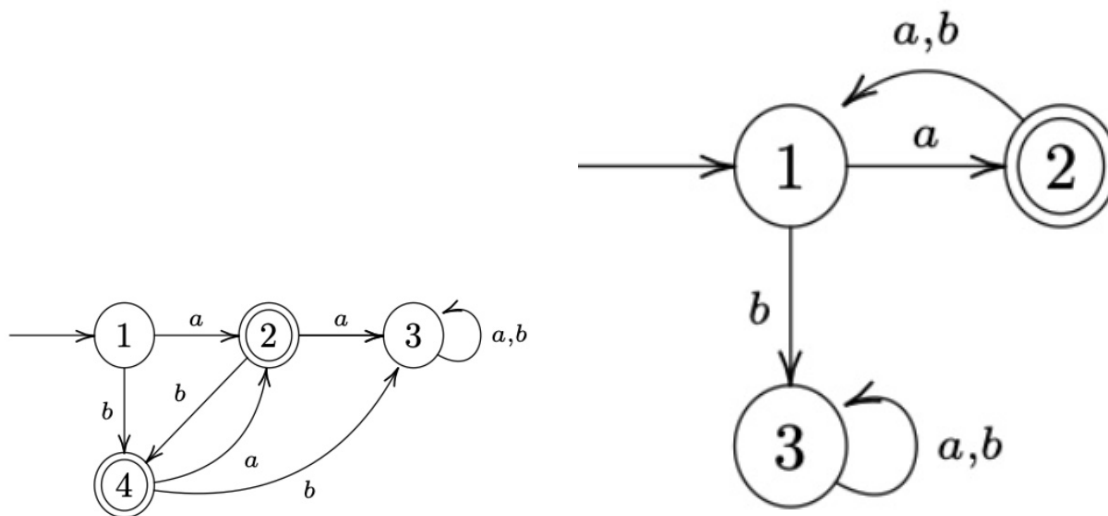


Figure 3: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

1. Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.

Answer:

- $L(\mathcal{A}^{(1)})$: Words include $a, b, ba, bab, baba, ab, aba, abab$, etc. I can see that each character must alternate based on its successive character.
- $L(\mathcal{A}^{(2)})$: Words include a, axa (where x doesn't matter). I see the word must start with a , and must add to its word length in increments of 2, every other character being a .

2. Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).

Answer: Below is the intersection automaton diagram.

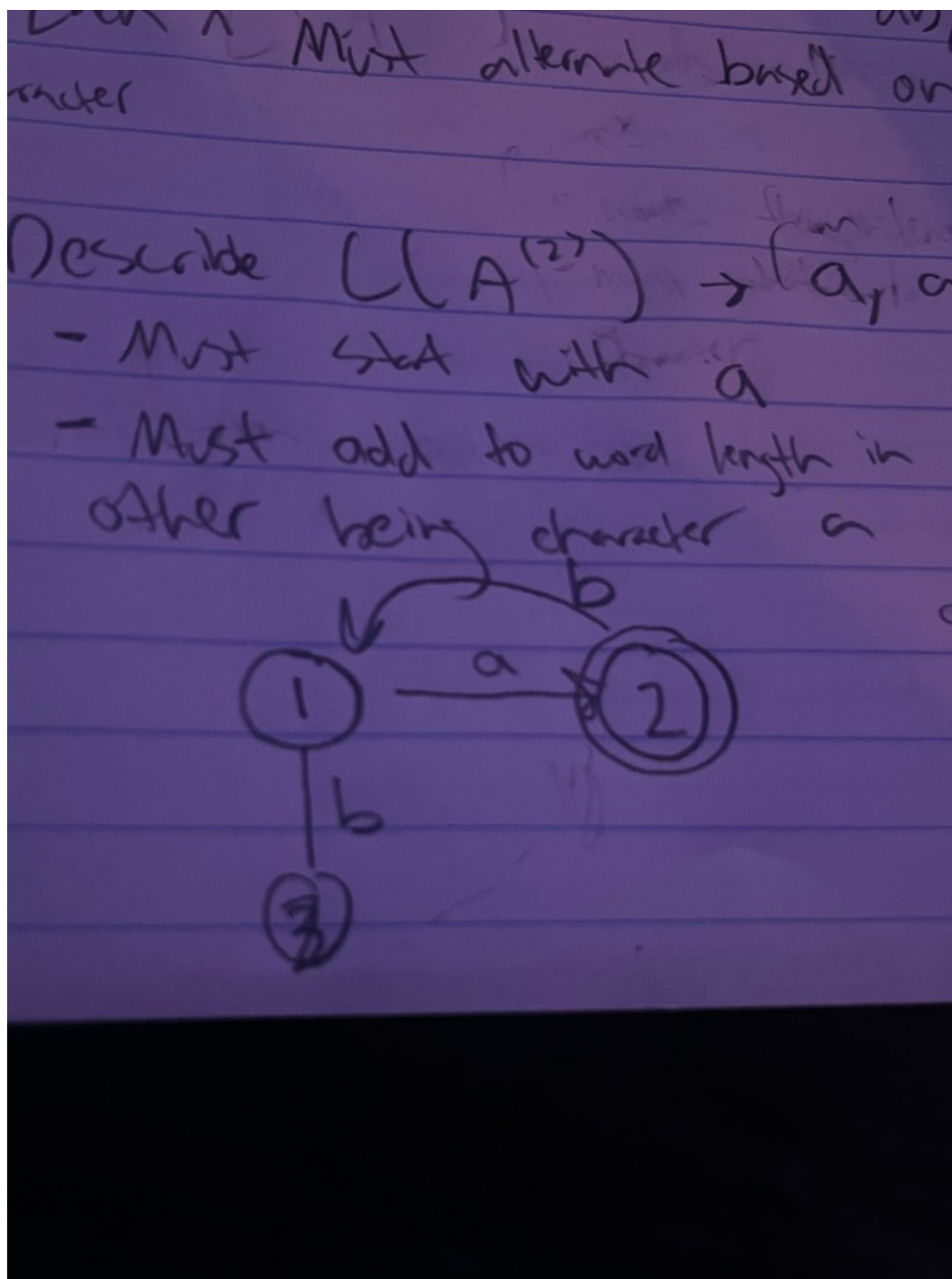


Figure 4: Intersection automaton $\mathcal{A}$ (hand-drawn diagram).

3. **Argue that** $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

Answer: Since $L(\mathcal{A}^{(1)})$ includes words such as a , b , ba , bab , $baba$ and ab , aba , $abab$, and $L(\mathcal{A}^{(2)})$ includes words that (1) must start with a —so we can disregard the words that start with b in $L(\mathcal{A}^{(1)})$,

leaving us with a, ab, aba , etc.—and (2) the requirement for $L(\mathcal{A}^{(2)})$ is to be in increments of two, we are left with patterns: $a, aba, ababa$, etc. Thus we obtain the automaton of both intersection.

4. **How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?**

Answer: We can allow for both word combinations to be accepting and add more state transitions.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.

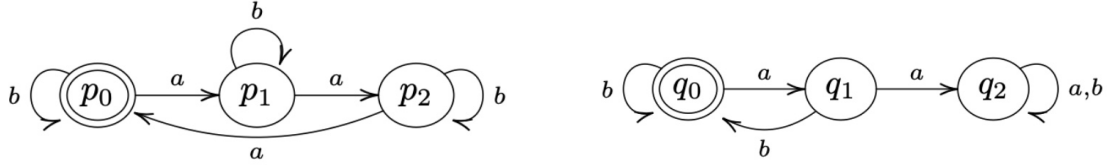


Figure 5: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\bar{L} := \Sigma^* \setminus L$.)

1. **Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.**

Answer:

Figure 6: $L(\mathcal{B}^{(1)})$ rules: include 3 a 's at a time, any amount of b 's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a , any amount of b 's. DFA diagram for language 2.

2. **Construct the intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).**

Answer: Below is the intersection automaton diagram.

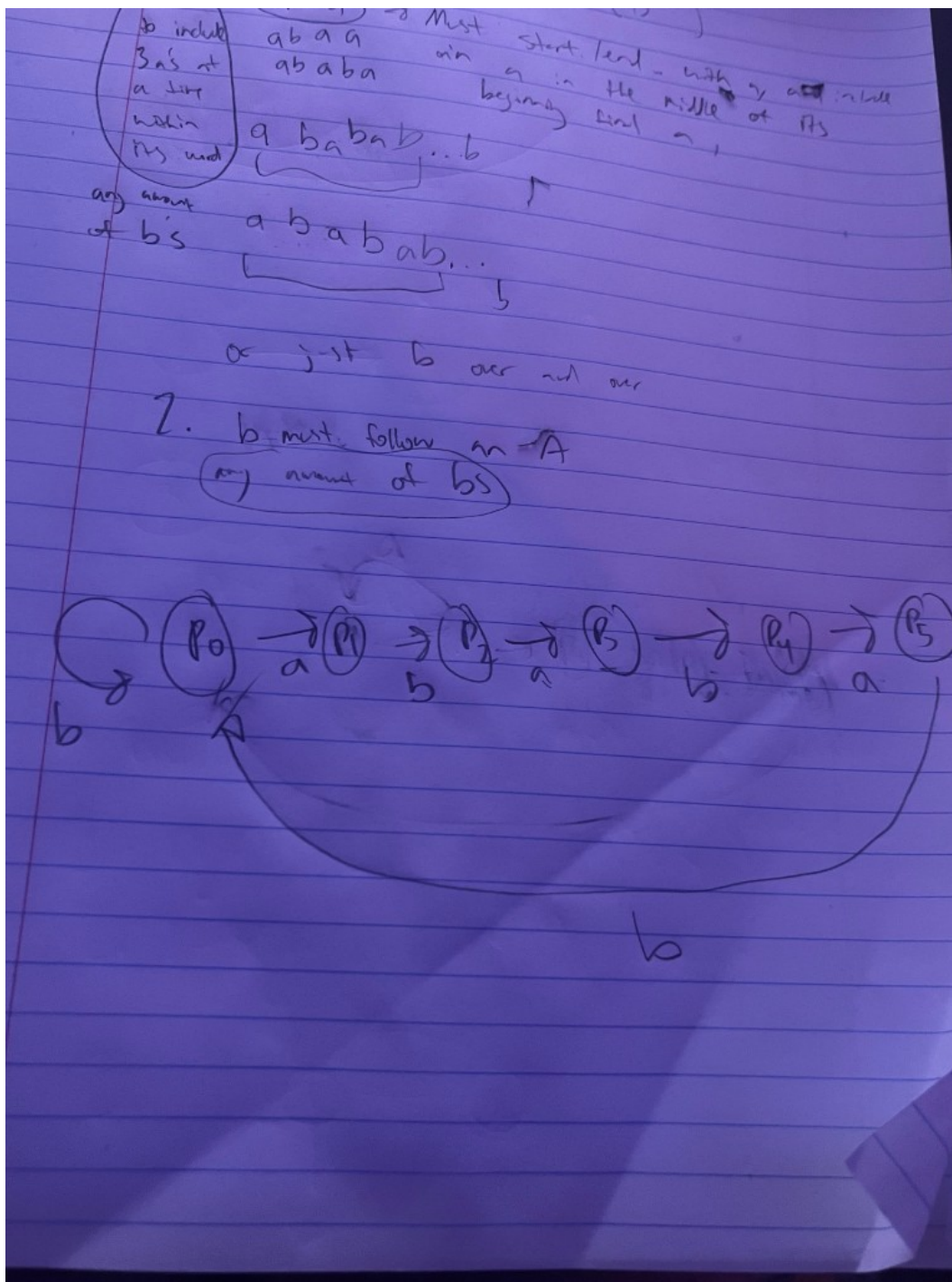


Figure 7: $L(\mathcal{B}^{(1)})$ rules: include 3 a 's at a time, any amount of b 's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a , any amount of b 's. DFA diagram for language 2.

3. **Argue that** $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$.

Answer: Both automata have the rule that any amount of b 's is allowed. But we must combine the

rules regarding a 's. $L(\mathcal{B}^{(1)})$ requires 3 a 's to be included at a time per word, while $\mathcal{B}^{(2)}$ requires the rule that a must be followed by a b . This gives us the conclusion that words that do include an a must come in pairs of 3 a 's with b following the last a .

4. **How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?**

Answer: We must flip the accepting states of automaton 1 to include: no amount of b 's AND words that include 3 a 's at a time only.

Exercise 3

Let $\mathcal{A}$ be a DFA and q a state of $\mathcal{A}$ such that from q , every input symbol a leaves us in q (i.e. $\delta(q, a) = q$ for all a). Show by induction on the length of the input that for any input string w , we still end up in q (i.e. $\delta^*(q, w) = q$).

Answer: We show it by induction on the length of w . (Here $\delta^*(q, w)$ is just the state we end up in after starting at q and reading the whole string w .)

When the input has length 0, the string is empty. So we don't read anything and we stay at q . So $\delta^*(q, w) = q$ in that case.

Now suppose we've already shown that for every string of length n , we end up back at q . Take a string w of length $n + 1$. We can write it as $w = va$ (some string v of length n plus one symbol a). After reading v we're in $\delta^*(q, v)$, and by our assumption that's q . Then we read a , and we're in $\delta(q, a)$, which is q by the given rule. So after reading w we're still in q , i.e. $\delta^*(q, w) = q$.

So by induction, for all input strings w we have $\delta^*(q, w) = q$.

Question

If you build the product automaton for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ but change the accepting set to get the *union* $L(\mathcal{B}^{(1)}) \cup L(\mathcal{B}^{(2)})$ instead of the intersection, how many states must be accepting?

2.3 Week 3

Homework 1 (NFA)

1. Describe the language accepted by $\mathcal{A}$ using a regular expression.
2. Specify the NFA $\mathcal{A}$ in the form $(Q, \Sigma, \delta, q_0, F)$.
3. Find all paths in $\mathcal{A}$ for the word $w = abab$; represent in a diagram.

Answer (NFA specification). Here is how I write the NFA $\mathcal{A}$ formally:

$$Q = \{0, 1, 2, 3\}, \quad \Sigma = \{a, b, c\}, \quad q_0 = 0, \quad F = \{2, 3\}.$$

For the transition function $\delta : Q \times \Sigma \rightarrow \mathcal{P}(Q)$ I use

$$\begin{aligned} \delta(0, a) &= \{0, 1\}, & \delta(0, b) &= \{0\}, & \delta(0, c) &= \emptyset, \\ \delta(1, b) &= \{2\}, & \delta(1, c) &= \{2\}, \end{aligned}$$

and any transition not listed goes to the empty set (i.e. there is no move).

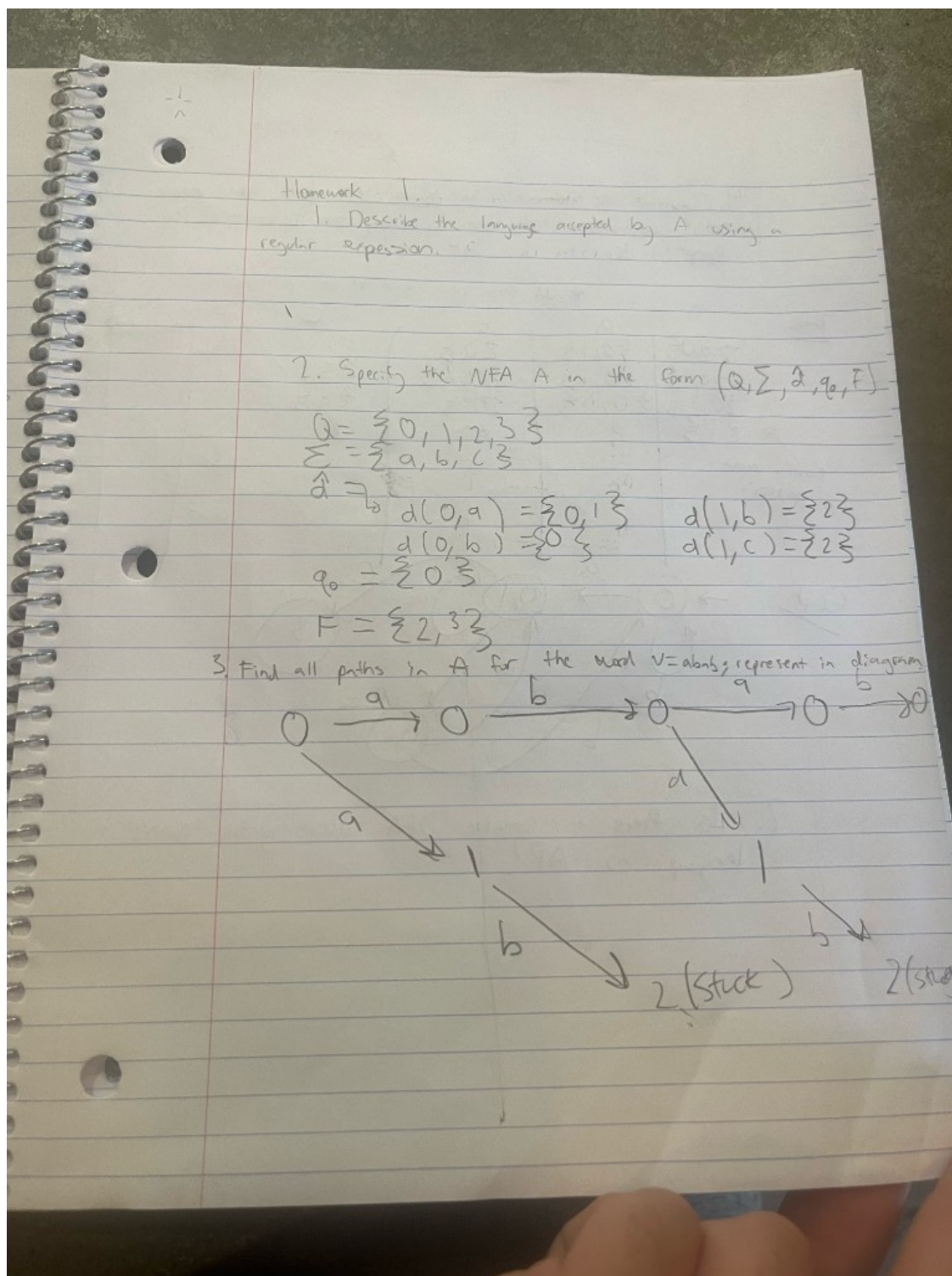


Figure 8: Homework 1: NFA specification $(Q = \{0, 1, 2, 3\}, \Sigma = \{a, b, c\}, q_0 = 0, F = \{2, 3\})$ and all paths for $w = abab$.

Homework 2 (Determinization)

Construct the determinization $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction. Create both a transition table and a graphical depiction of $\mathcal{A}^D$.

Transition table for $\mathcal{A}^D$ over $\{a, b, c\}$. Starting from the NFA above, the reachable subset-states under the power set construction are

$$\{0\}, \{0, 1\}, \{0, 1, 2\}, \{0, 2, 3\}, \{2, 3\}, \emptyset.$$

I call a subset accepting when it contains either 2 or 3.

State S	on input a	on input b	on input c
$\{0\}$	$\{0, 1\}$	$\{0\}$	$\emptyset$
$\{0, 1\}$	$\{0, 1\}$	$\{0, 2\}$	$\{2\}$
$\{0, 1, 2\}$	$\{0, 1\}$	$\{0, 2\}$	$\{2\}$
$\{0, 2\}$	$\{0, 1\}$	$\{0, 2\}$	$\emptyset$
$\{2\}$	$\emptyset$	$\{2\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

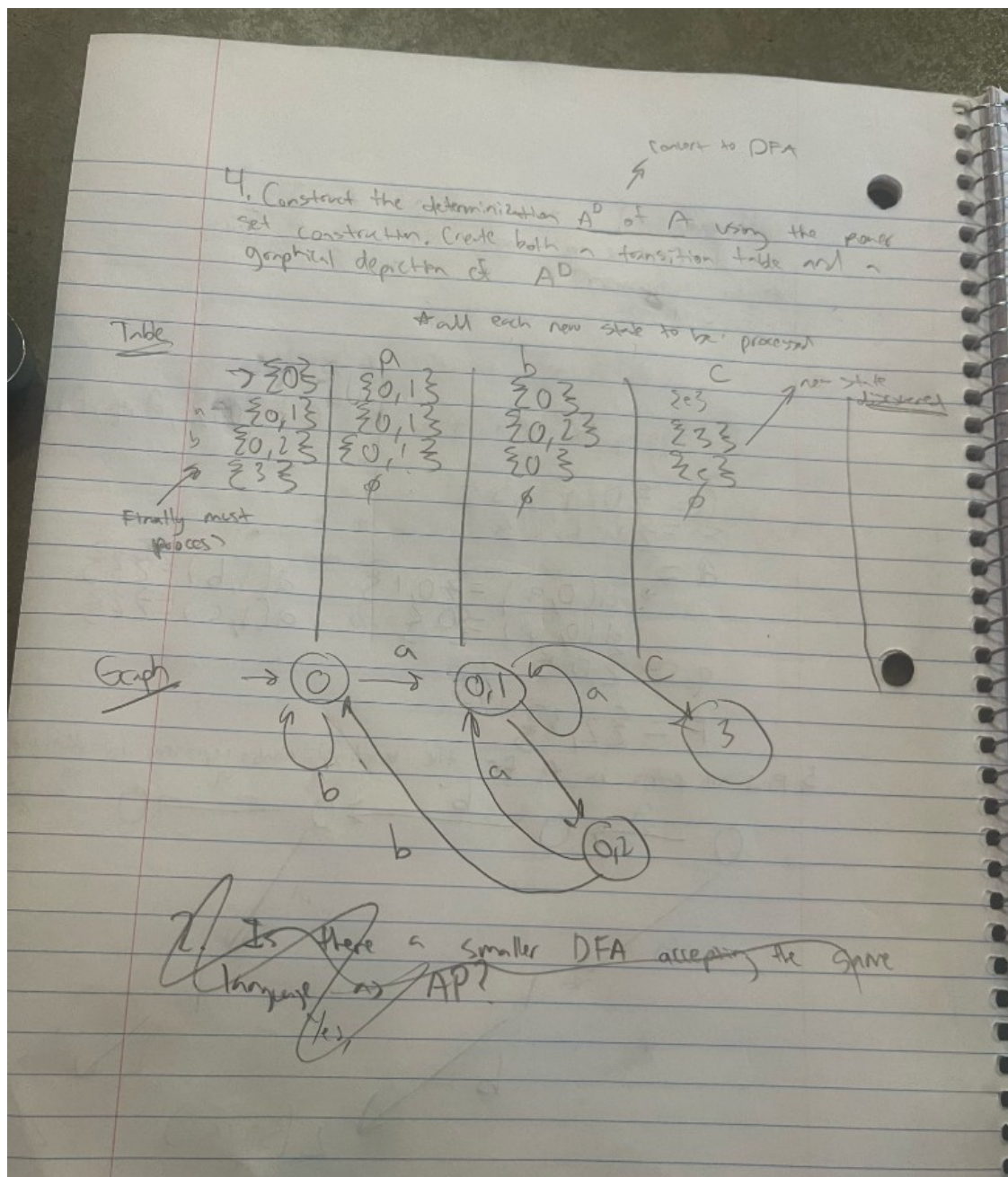


Figure 9: Graph of A^D (photo); the transition table is typeset separately in LaTeX.

Homework 3 (DFA and minimization)

Determine A^D of A using the power set construction, now with alphabet $\{0,1\}$. Create both a transition table and a graphical depiction of A^D . Then decide whether there can be a smaller DFA accepting the same language.

Transition table over $\{0, 1\}$. From the NFA given in the homework, the reachable subset-states are

$$\{q_0\}, \{q_0, q_1\}, \{q_0, q_1, q_2\}, \{q_0, q_1, q_2, q_3\}, \{q_0, q_3\}.$$

The determinized DFA $\mathcal{A}^D$ has the following transitions:

State S	on input 0	on input 1
$\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1\}$
$\{q_0, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1\}$

Any subset that contains the original accepting state(s) is accepting in $\mathcal{A}^D$.

Smaller DFA? Yes. In this table the last two rows (for $\{q_0, q_1, q_2, q_3\}$ and for $\{q_0, q_3\}$) have exactly the same outgoing transitions and the same accepting/non-accepting status. That means these two states are equivalent and can be merged into a single state, giving a smaller DFA that still recognizes the same language.

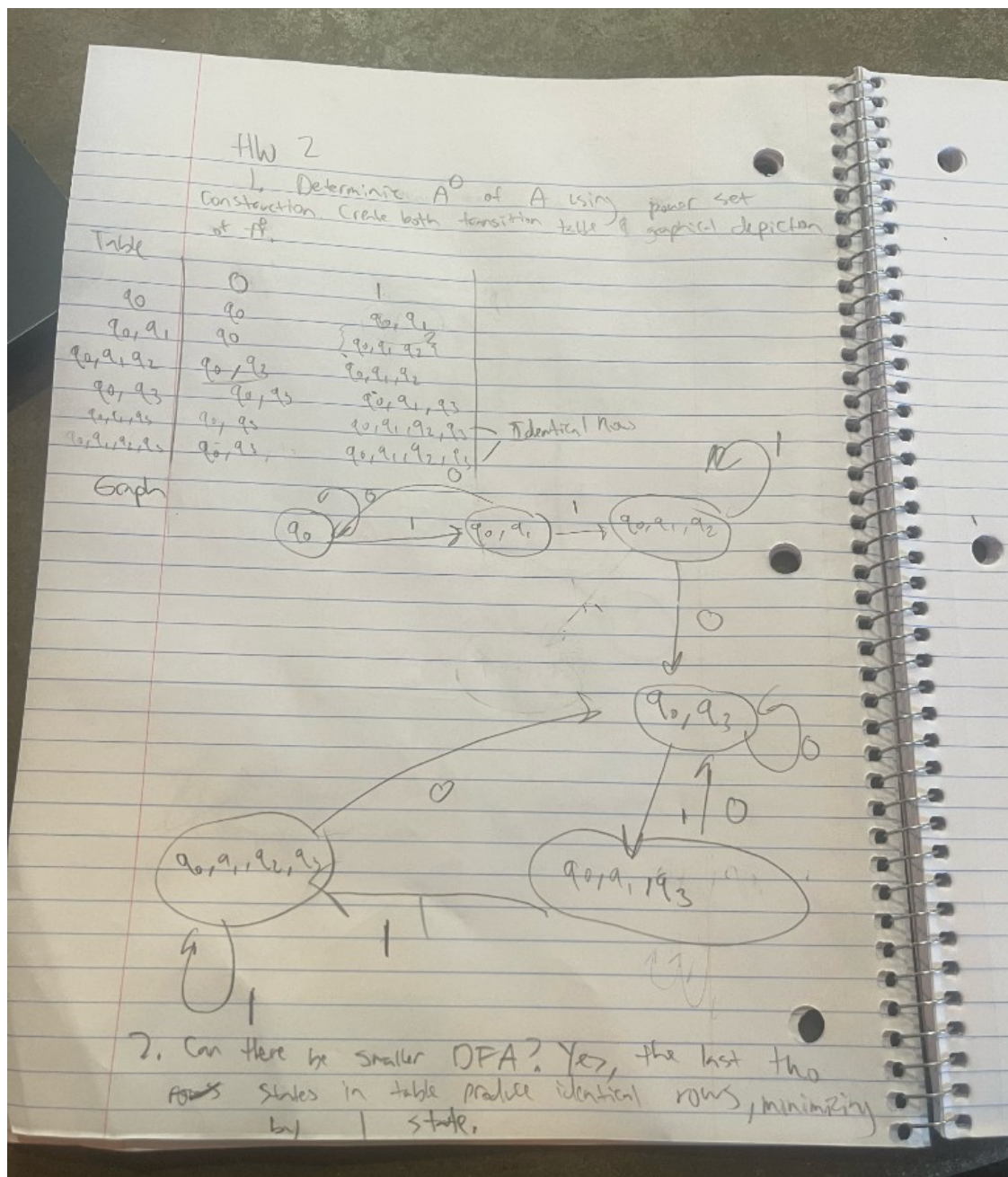


Figure 10: State diagram of A^D for Homework 3 (photo of the graph only).

Question

After building A^D from an NFA A using the power set construction, when can we get a smaller DFA that still accepts the same language?

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.